

Reset State

POST /api/reset

Request:

```
{  
  N/A  
}
```

Response:

```
{  
  "status": "reset"  
}
```

Side Effects:

- Clears all game data from the database
 - Resets all active games
-

Create Player

POST /api/players

Request:

```
{  
  "username": "johnDoe"  
}
```

Response:

```
{  
  "player_id": 1  
}
```

Side Effects:

- Creates a new player record in the players table
 - Initializes player statistics (wins, losses, games played)
 - Generates a unique player_id
-

Retrieve Player Statistics

GET /api/players/{id}/stats

Request:

```
{
  N/A
}
```

Response:

```
{
  "games_played": 3,
  "wins": 1,
  "losses": 2,
  "total_shots": 24,
  "total_hits": 9,
  "accuracy": 0.375
}
```

Side Effects:

- Will pull player stats to the frontend for display

Create New Game

POST /api/games

Request:

```
{
  "creator_id": 1,
  "grid_size": 10,
  "max_players": 2
}
```

Response:

```
{
  "game_id": 10,
  "grid_size": 10,
  "status": "waiting",
  "current_turn_index": 0
}
```

Side Effects:

- Creates a new game record in the Games table
- Adds creator to GamePlayers
- Initializes turn order
- Sets game status to waiting

Constraints:

- 1:N max_players
 - 5:15 grid_size
-

Join Existing Game

POST /api/games/{id}/join

Request:

```
{
  "player_id": 2
}
```

Response:

```
{
  "status": "joined"
}
```

Side Effects:

- Adds a player to the GamePlayers table
 - Assigns a turn order position
 - Updates the player count for the game
 - If maximum players join the session game status changes to active
-

Retrieve Game State

GET /api/games/{id}

Request:

```
{  
  N/A  
}
```

Response:

```
{  
  "game_id": 10,  
  "grid_size": 8,  
  "status": "active",  
  "current_turn_index": 1,  
  "active_players": 2  
}
```

Side Effects:

- Retrieves current game progress
-

Place Ships

POST /api/games/{id}/place

Request:

```
{  
  "player_id": 1,  
  "ships": [  
    { "row": 0, "col": 0 },  
    { "row": 1, "col": 0 },  
    { "row": 2, "col": 0 }  
  ]  
}
```

Response:

```
{  
  "status": "ships_placed"  
}
```

Side Effects:

- Stores the ship coordinates for each player
- Validates the ship placement rules
- Marks the player as ready

Constraints:

- Exactly 3 single-cell ships per player per project requirements
 - Ships cannot be placed outside the grid
 - Ships cannot overlap
 - The same ship cannot be placed twice
 - All ships must be placed before the game starts
-

Fire Shot at Opponent

POST /api/games/{id}/fire

Request:

```
{
  "player_id": 2,
  "row": 3,
  "col": 4
}
```

Response:

```
{
  "result": "hit",
  "next_player_id": 3,
  "game_status": "active"
}
```

If no more opponent ships are left after the shot

```
{
  "result": "hit",
  "next_player_id": null,
  "game_status": "finished",
  "winner_id": 2
}
```

Side Effects:

- Records the shot in the Moves table
- Determines hit or miss
- Updates the player's statistics

- Advances the turn order
 - Checks if a player has lost all of their ships
 - If only one player remains, the game status is changed to finished.
-

Retrieve Move History

GET /api/games/{id}/moves

Request:

```
{  
    N/A  
}
```

Response:

```
{  
    "moves": [  
        {  
            "player_id": 2,  
            "row": 3,  
            "col": 4,  
            "result": "hit",  
            "timestamp": "2026-03-06T12:00:00Z"  
        }  
    ]  
}
```

Side Effects:

- Returns the history of all moves in a game
-

Restart Game (Used for grading during testing)

POST /api/test/games/{id}/restart

Request:

```
{
  N/A
}
```

Response:

```
{
  "status": "game_restarted"
}
```

Side Effects:

- Clears all ships from the current game
 - Clears all recorded moves
 - Resets the game status to waiting
 - Player statistics are unchanged
-

Deterministic Ship Placement (used for grading during testing)

POST /api/test/games/{id}/ships

Request:

```
{
  "player_id": 1,
  "ships": [
    { "row": 0, "col": 0 },
    { "row": 1, "col": 0 },
    { "row": 2, "col": 0 }
  ]
}
```

Response:

```
{
  "status": "ships_set"
}
```

Side Effects:

- Sets ship coordinates deterministically
- Overrides normal ship placement validation (via player)

Reveal Board State (used for grading during testing)

POST /api/test/games/{id}/board/{player_id}

Request:

```
{  
  N/A  
}
```

Response:

```
{  
  "ships": [  
    { "row": 0, "col": 0 },  
    { "row": 1, "col": 0 },  
    { "row": 2, "col": 0 }  
  ],  
  "moves": [  
    { "row": 3, "col": 4, "result": "hit" }  
  ]  
}
```

Side Effects:

- Reveals the full board state